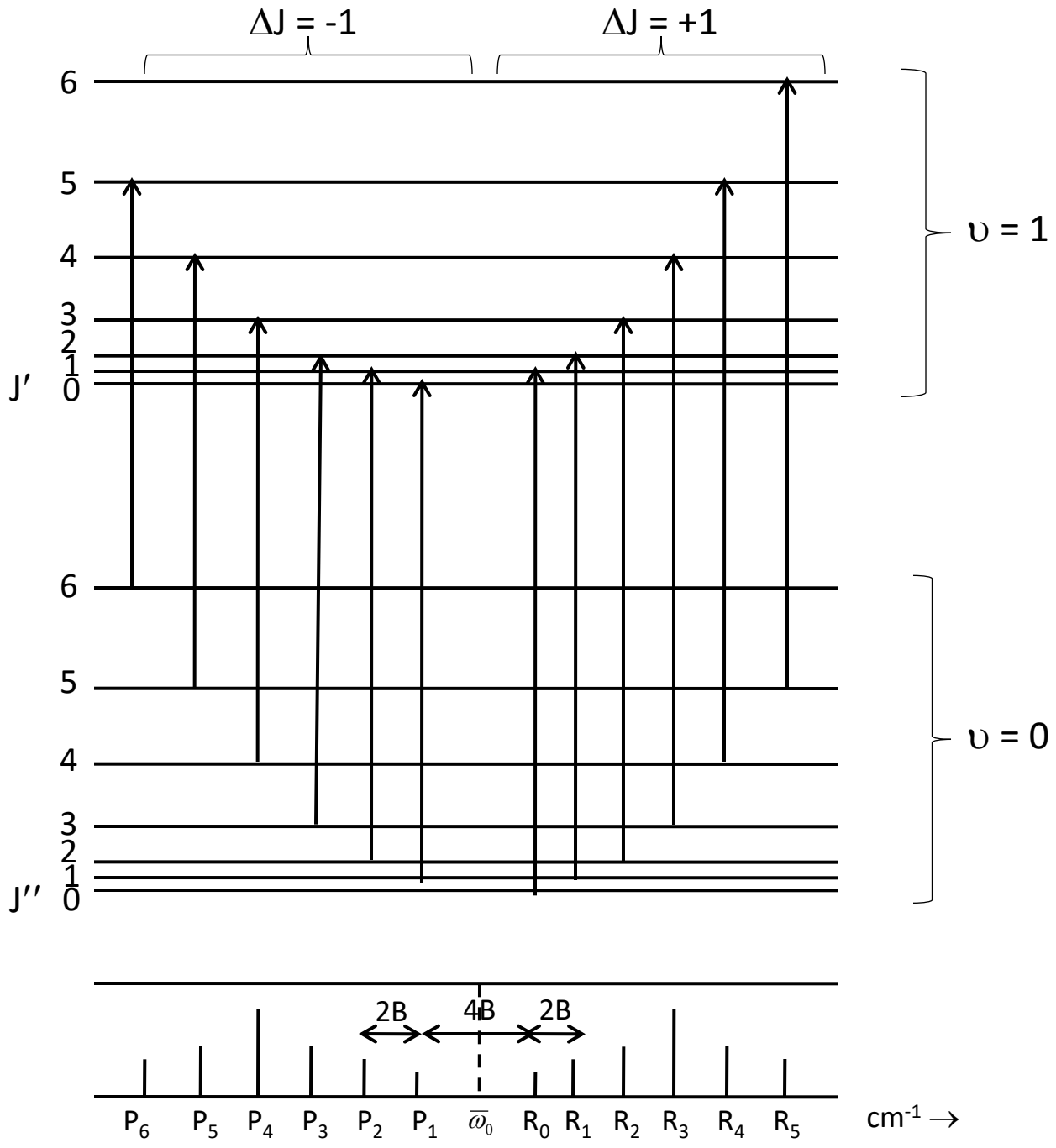



v(0), J(0) → v(1), J(1)
Δv=+1, ΔJ=+1

v(0), J(1) → v(1), J(0)
Δv=+1, ΔJ=-1

Filled to varying degrees in molecular population (Boltzmann distribution).

An IR spectrometer can record either the absorption or the emission spectrum.

# The influence of Nuclear Spin

Example: $CO_2$

Centre of symmetry in a molecule affects the intensity of alternate lines in the P and R branches.

This is due to the existence of **nuclear spin**, which is an additional factor in determining the population of the rotational levels.

Classical theory states that at a temperature T K, the **ratio of the population** of such levels will be given by:

$$\frac{N_{upper}}{N_{lower}} = exp\left(-\frac{\Delta E}{kT}\right)$$

where k is the Boltzmann constant.

Thus, at all temperatures above absolute zero the upper level will always be populated to some extent.

$O^{16}=C=O^{16}$

Symmetric molecule.

- Every alternate rotational level is completely unoccupied.
- Alternate lines in the P and R branches have zero intensity.
- The line spacing in the spectrum is 4B instead of 2B.

$O^{18}=C=O^{16}$

No centre of symmetry.

The line spacing in the spectrum is 2B.

With a nuclear spin of zero meaning, it is a **boson** - it's total wavefunction must be totally symmetric.

The **total wavefunction** is made up of an electronic, vibrational, rotational and a nuclear spin contributions. $\Psi_{total} = \psi_{elec}\psi_{vib}\psi_{rot}\psi_{ns}$

For this molecule the electronic and vibrational (in ground state) wavefunctions are symmetrical.

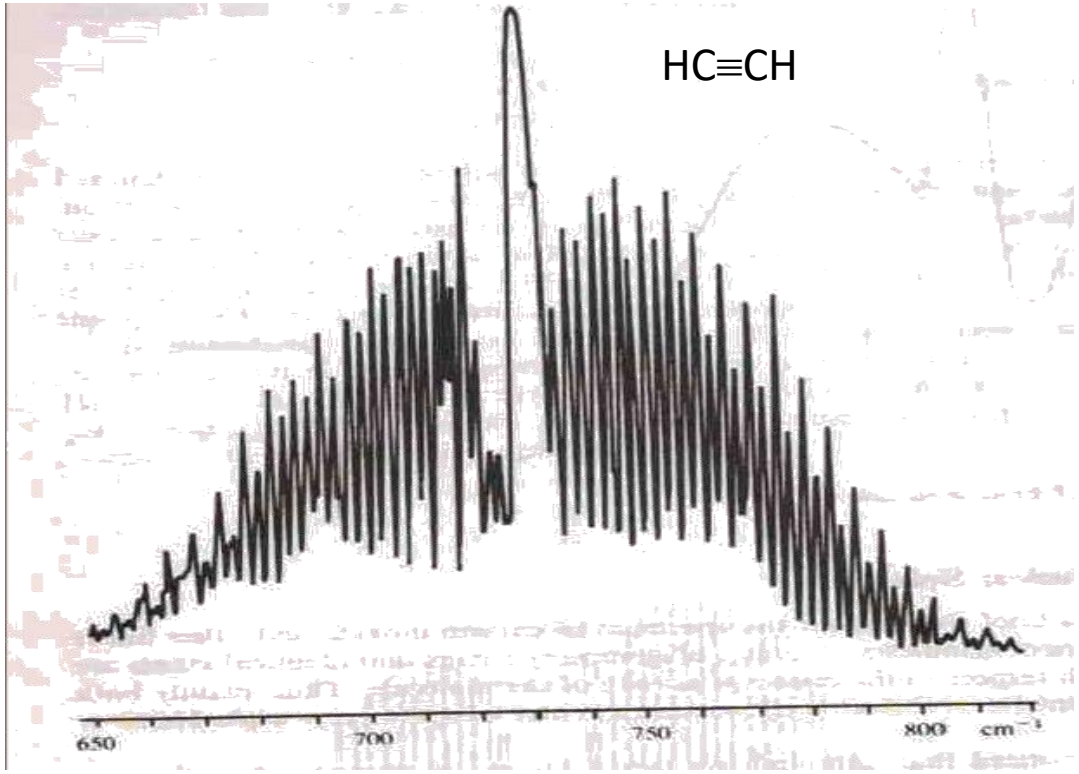
There is no antisymmetric nuclear spin wavefunction <u>so only the even J rotational levels exist</u>.

For **fermions**, the total wavefunction changes sign and is antisymmetric when two identical particles are exchanged, whereas for **bosons** the wavefunction does not change sign and is symmetric.

# Compare HC≡CH and DC≡CD

Alternate levels have population ratio of 3:1.

The P and R branch lines show a strong-weak-strong…. Signal due to nuclear spin of the hydrogen atoms.




Nuclei, such as H, that have a nuclear spin angular momentum quantum number of ½, are fermions,

whereas nuclei such as D with a spin quantum number of 1, are bosons.

Because of nuclear spin statistics, the relative intensities of the rotational lines in the IR absorption spectrum of $C_2H_2$ and $C_2D_2$ are qualitatively different.

For $C_2HD$, the two hydrogen atoms are not identical, so nuclear spin statistics do not affect the intensities.

Although the carbon atoms are identical, $^{12}C$ has a nuclear spin of zero, which means the degeneracy of the nuclear spin wavefunction is one, which in turn means there is <u>no effect on the relative intensities</u>.

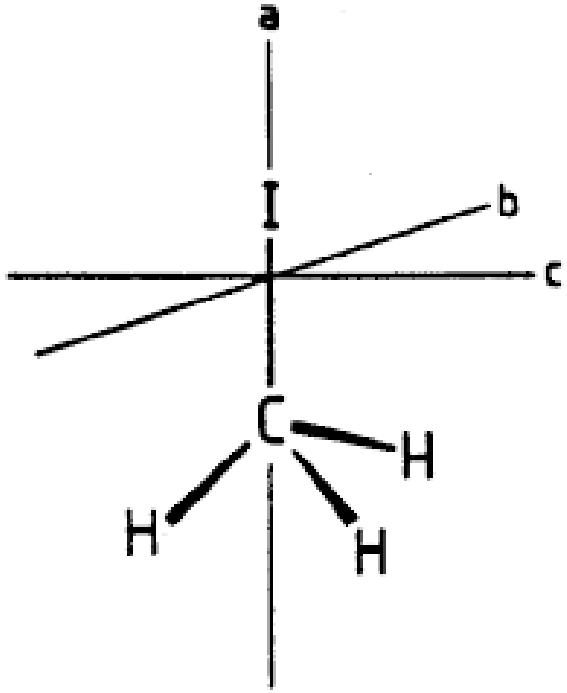
# Symmetric Top Molecules

Adopting Born-Oppenheimer approximation:

$$\varepsilon_{J,\upsilon} = \varepsilon_{vib} + \varepsilon_{rot}$$

$$= \left(\upsilon + \frac{1}{2}\right)\overline{\omega}_e - \left(\upsilon + \frac{1}{2}\right)^2 x_e\overline{\omega}_e + BJ(J+1) + (A-B)K^2 \ \ \text{cm}^{-1}$$

Assuming negligible centrifugal distortion.



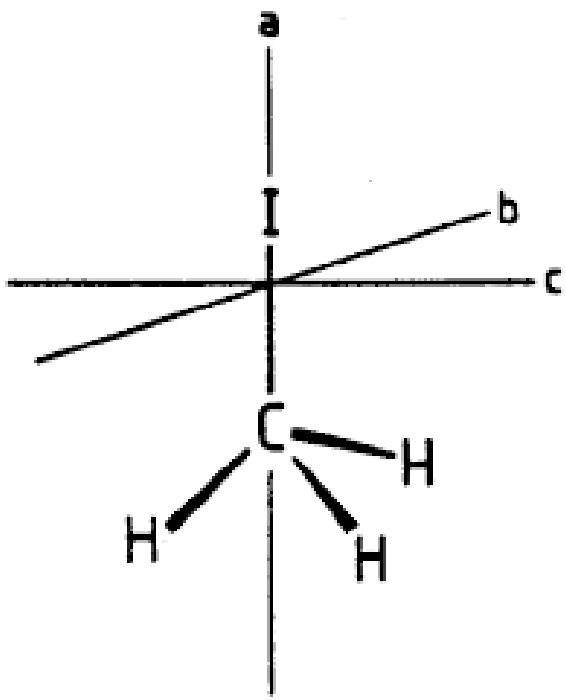

In a symmetric top molecule,

- "parallel vibrations" refer to vibrational modes where the oscillating dipole moment is aligned with the molecule's symmetry axis,
- "perpendicular vibrations" are those where the dipole moment oscillates perpendicular to the symmetry axis

J $\rightarrow$ end-over-end rotation and K $\rightarrow$ about the top axis (change in the angular momentum)

# Symmetric Top Molecules

## Parallel vibrations




**Selection Rules:**

$\Delta J = 0, \pm 1; \Delta \upsilon = \pm 1; \Delta K = 0$

The spectral frequency is independent of K.

The spectrum contains P, Q and R branches.

P, R line spacing is 2B and a strong Q branch.

The intensity of Q branch varies with the ratio $I_A/I_B$.

When $I_A$ tends to zero, the symmetric top becomes linear. Q branch has zero intensity.

$I_A \rightarrow$ rotation about the main symmetry axis
$I_B$, $I_C \rightarrow$ rotation perpendicular to the main axis
$I_B = I_C \neq I_A$ and $I_A \neq 0$

# Symmetric Top Molecules
# Perpendicular vibrations

**Selection Rules:**

$\Delta J = 0, \pm 1; \Delta \upsilon = \pm 1; \Delta K = \pm 1$

$\Delta J = +1; \Delta K = \pm 1 \rightarrow$ R branch lines.

$\Delta J = -1; \Delta K = \pm 1 \rightarrow$ P branch lines.

$\Delta J = 0; \Delta K = \pm 1 \rightarrow$ Q branch lines.

## Linear molecules:

### Parallel Vibrations

$\Delta J = \pm 1$, $\Delta \upsilon = \pm 1$ (for simple harmonic oscillators)
$\Delta J = \pm 1$, $\Delta \upsilon = \pm 1, \pm 2, \pm 3, \ldots$ (for anharmonic oscillators)

### Perpendicular Vibrations

$\Delta J = 0, \pm 1$, $\Delta \upsilon = \pm 1$ (for simple harmonic oscillators)

## Symmetric Top Molecules:

### Parallel Vibrations

$\Delta J = 0$, $\Delta \upsilon = \pm 1$, $\Delta K = 0$

### Perpendicular Vibrations

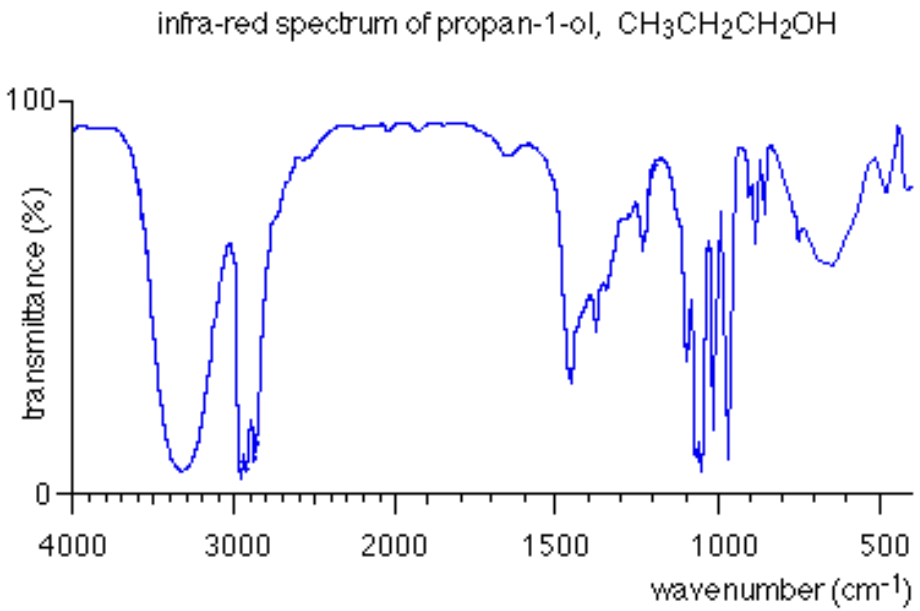
$\Delta J = 0, \pm 1$; $\Delta \upsilon = \pm 1$; $\Delta K = \pm 1$

# Infra-red Spectrum

The normal modes of vibrations are divided into:

(i) Skeletal vibrations
(ii) Group vibrations

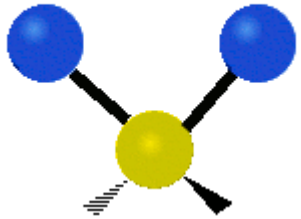
## What an infra-red spectrum looks like



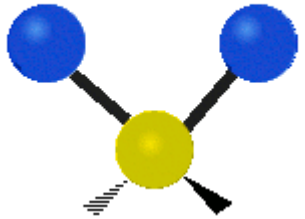
**What causes some frequencies to be absorbed?**

Each frequency of light (including infra-red) has a certain energy. If a particular frequency is being absorbed as it passes through the compound being investigated, it must mean that its energy is being transferred to the compound.

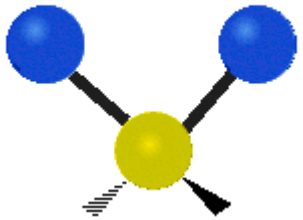
Energies in infra-red radiation correspond to the energies involved in bond vibrations.



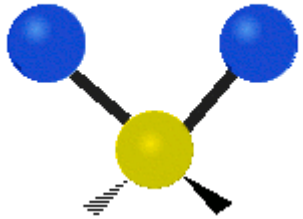
Symmetric stretch



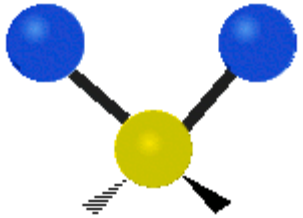
Anti-symmetric stretch



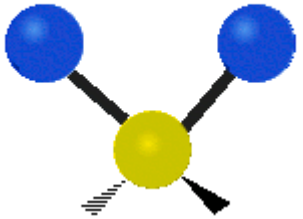
Wagging



Twisting



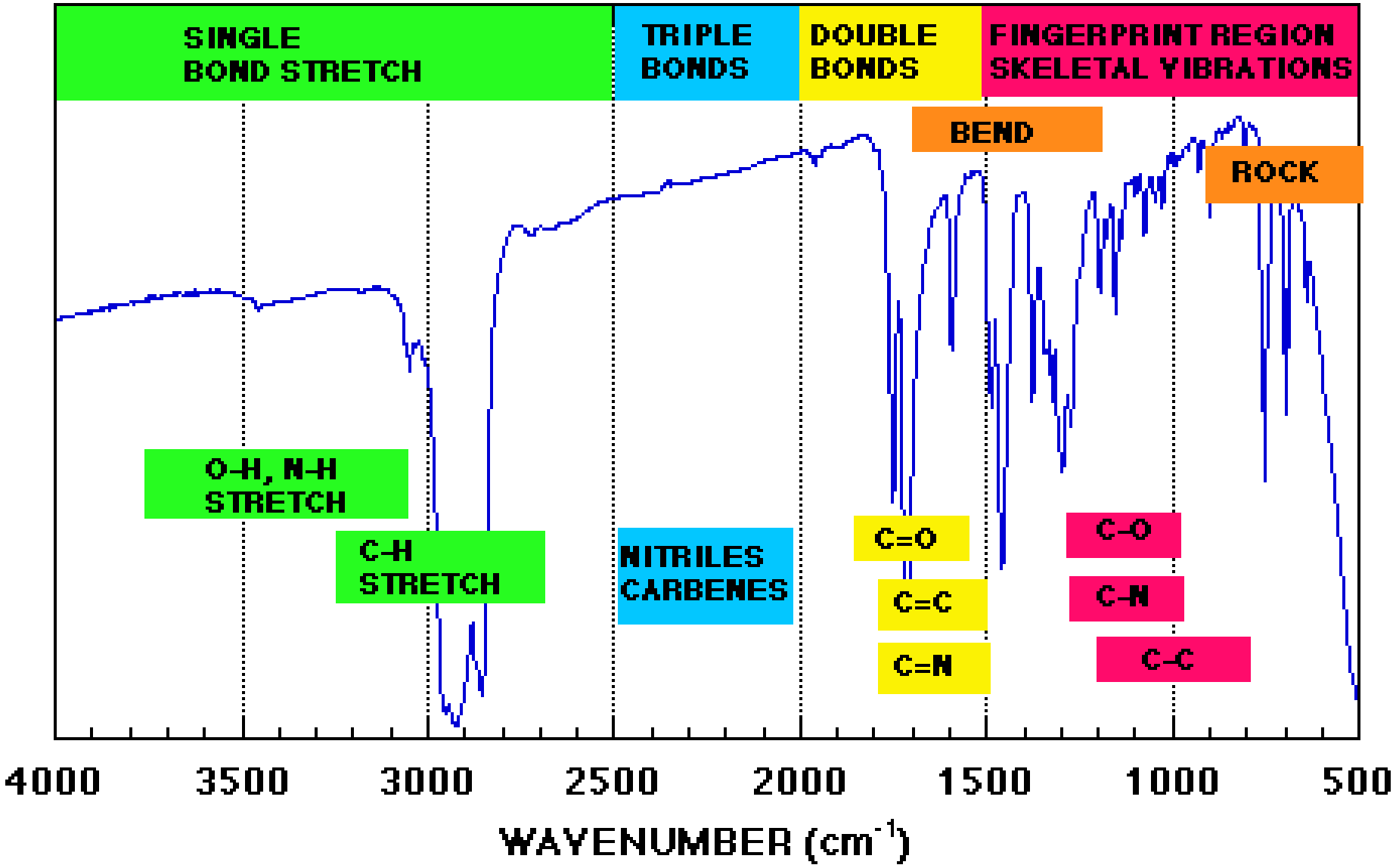
Scissoring



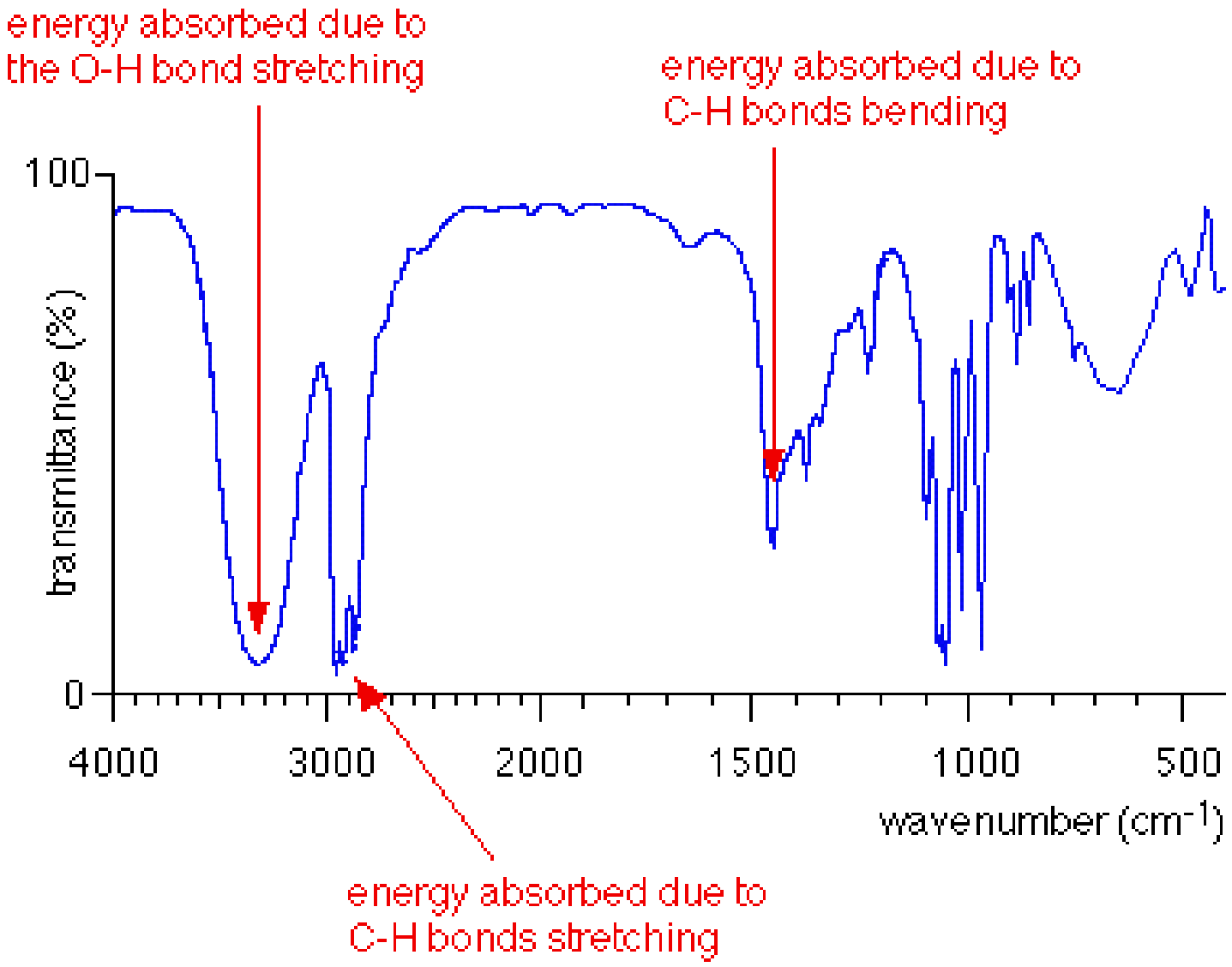
Rocking

Look for absorption bands in decreasing order of importance:

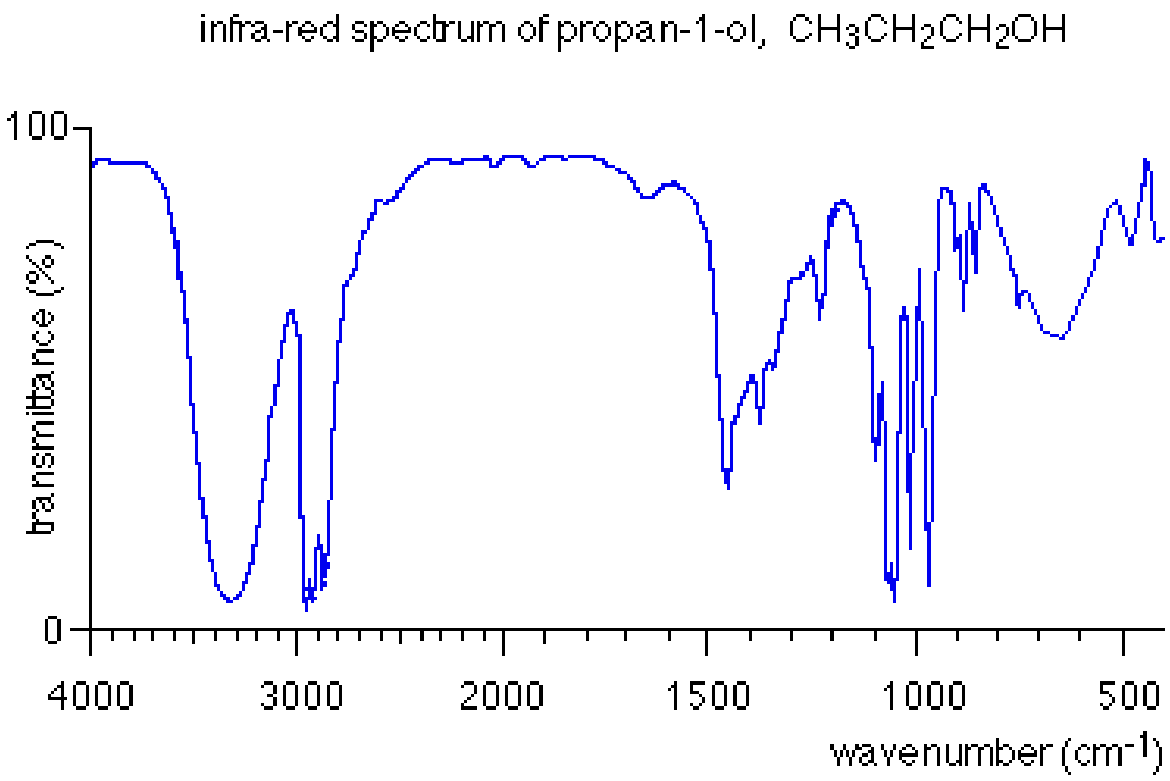
1. the **C-H** absorption(s) between **3100 and 2850 cm$^{-1}$**. An absorption above **3000 cm$^{-1}$** indicates **C=C**, either alkene or aromatic. Confirm the **aromatic ring** by finding peaks at **1600 and 1500 cm$^{-1}$** and **C-H out-of-plane bending** to give substitution patterns **below 900 cm$^{-1}$**. Confirm **alkenes** with an absorption at **1640-1680 cm$^{-1}$**. **C-H** absorption between **3000** and **2850 cm$^{-1}$** is due to **aliphatic hydrogens**.
2. the carbonyl (**C=O**) absorption between **1690-1760 cm$^{-1}$**; this strong band indicates either an aldehyde, ketone, carboxylic acid, ester, amide, anhydride or acyl halide. An **aldehyde** may be confirmed with **C-H absorption from 2840 to 2720 cm$^{-1}$**.
3. the **O-H or N-H** absorption between **3200 and 3600 cm$^{-1}$**. This indicates either an alcohol, N-H containing amine or amide, or carboxylic acid. For **-NH$_2$ a doublet** will be observed.
4. the **C-O** absorption between **1080 and 1300 cm$^{-1}$**. These peaks are normally rounded like the O-H and N-H peak in 3 and are prominent. Carboxylic acids, esters, ethers, alcohols and anhydrides all contain this peak.
5. the **CC and CN triple bond** absorptions at **2100-2260 cm$^{-1}$** are small but exposed.
6. a **methyl group** may be identified with **C-H absorption at 1380 cm$^{-1}$**. This band is split into a **doublet for isopropyl (*gem*-dimethyl) groups**.
7. structure of **aromatic compounds** may also be confirmed from the pattern of the **weak overtone and combination tone bands found from 2000 to 1600 cm$^{-1}$**.

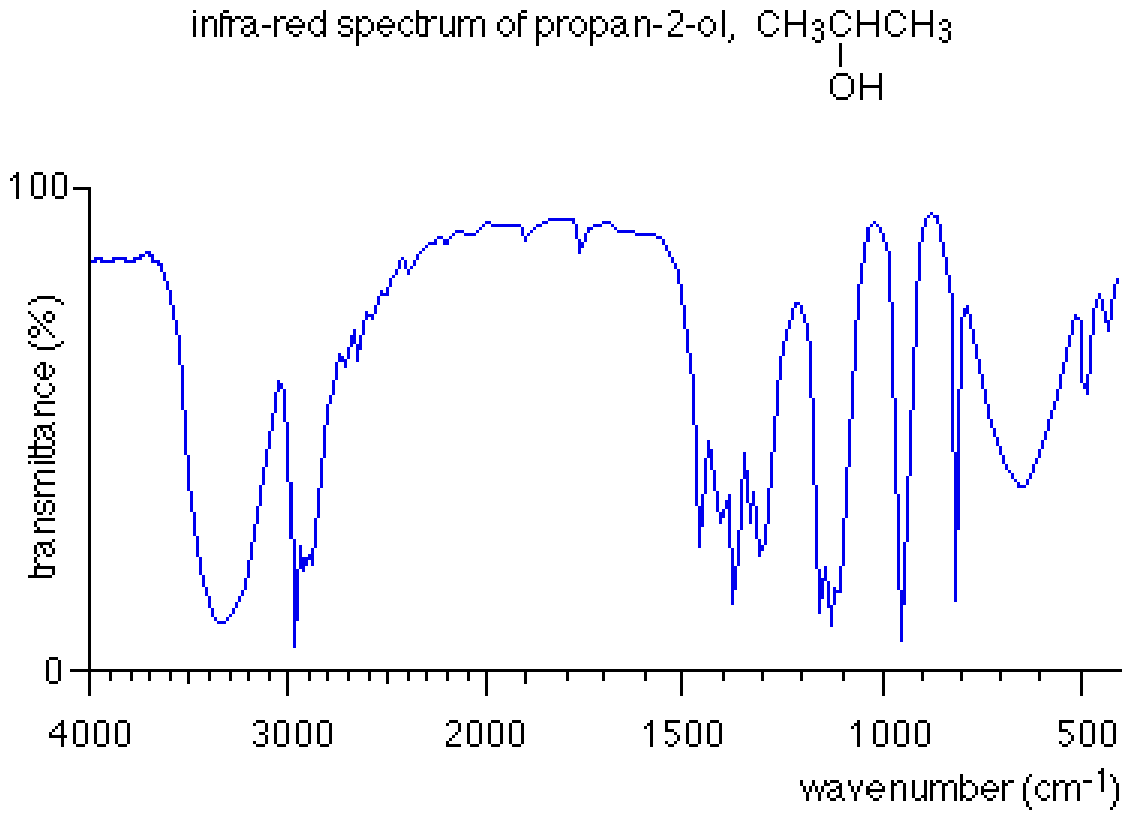

SINGLE BOND STRETCH
TRIPLE BONDS
DOUBLE BONDS
FINGERPRINT REGION SKELETAL VIBRATIONS
BEND
ROCK
O-H, N-H STRETCH
C-H STRETCH
NITRILES CARBENES
C=O
C=C
C=N
C-O
C-N
C-C
4000
3500
3000
2500
2000
1500
1000
500
WAVENUMBER (cm-1)

# Infra-red spectrum of propan-1-ol, $CH_3CH_2CH_2OH$



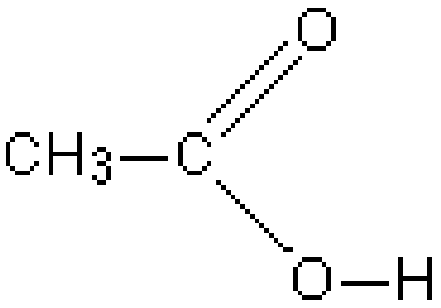
# Compare the two spectra



**Ethanoic acid**

Ethanoic acid has the structure:




It contains the following bonds:

carbon-oxygen double, C=O
carbon-oxygen single, C-O
oxygen-hydrogen, O-H
carbon-hydrogen, C-H
carbon-carbon single, C-C

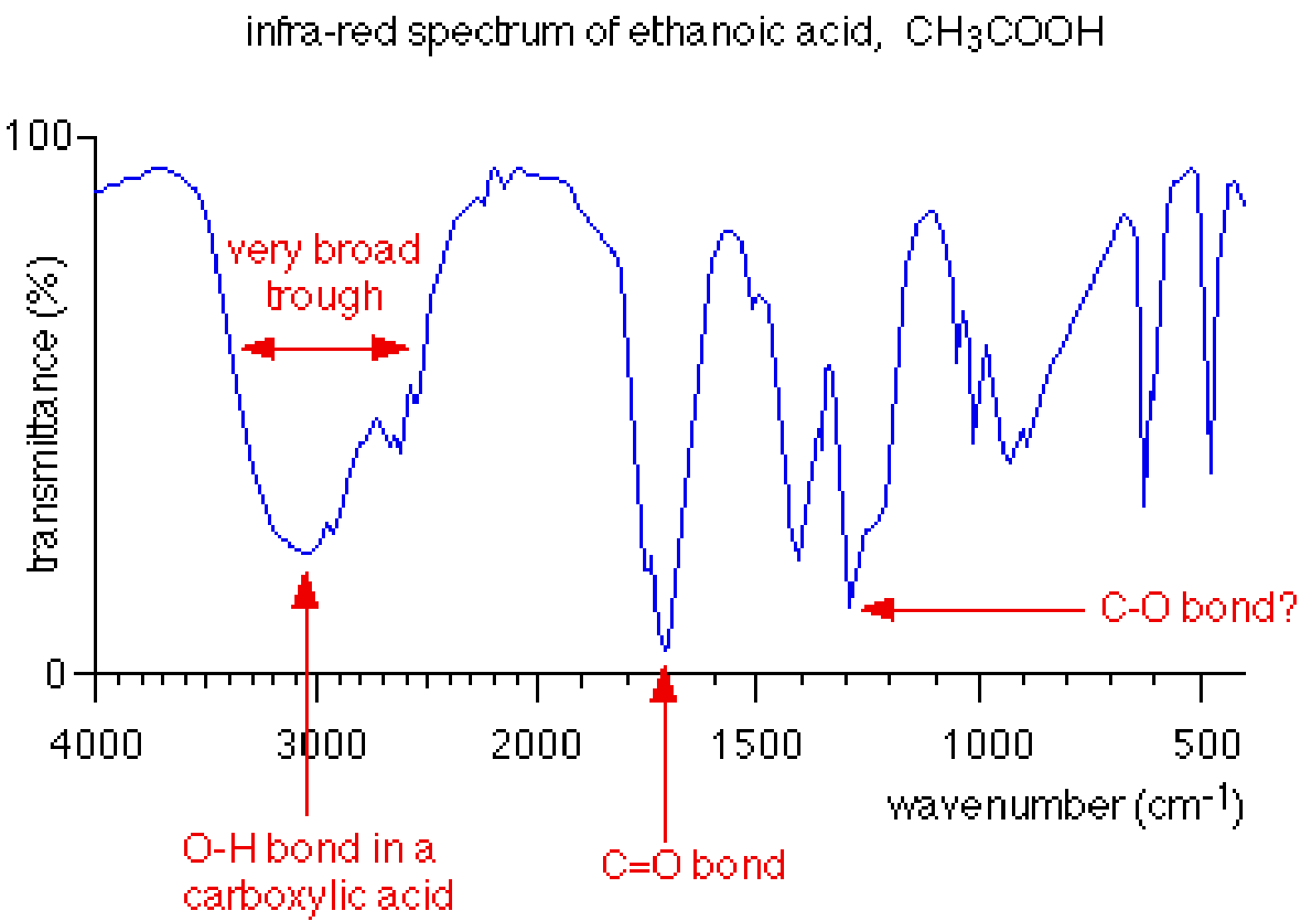
The carbon-oxygen double bond, C=O is in the range 1680 - 1750 $cm^{-1}$.

Its position varies slightly depending on what sort of compound it is in.

The other useful bond is the O-H bond.

This absorbs differently depending on its environment.

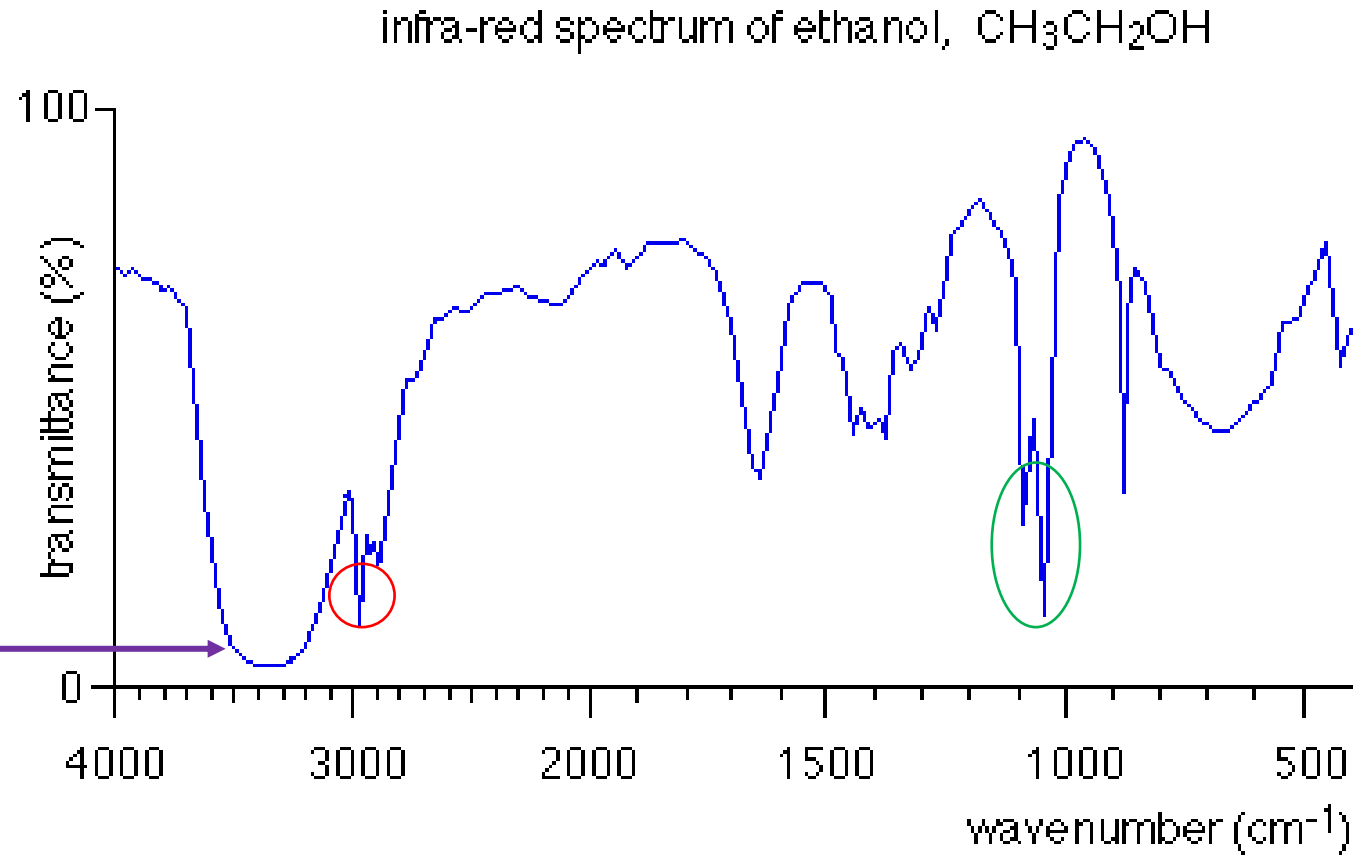
It is easily recognised in an acid because it produces a very broad trough in the range 2500 - 3300 $cm^{-1}$.


infra-red spectrum of ethanoic acid, CH3COOH
100
0
transmittance (%)
very broad trough
C-O bond?
4000
3000
2000
1500
1000
500
wavenumber (cm-1)
O-H bond in a carboxylic acid
C=O bond

# The infra-red spectrum for an alcohol

## Ethanol

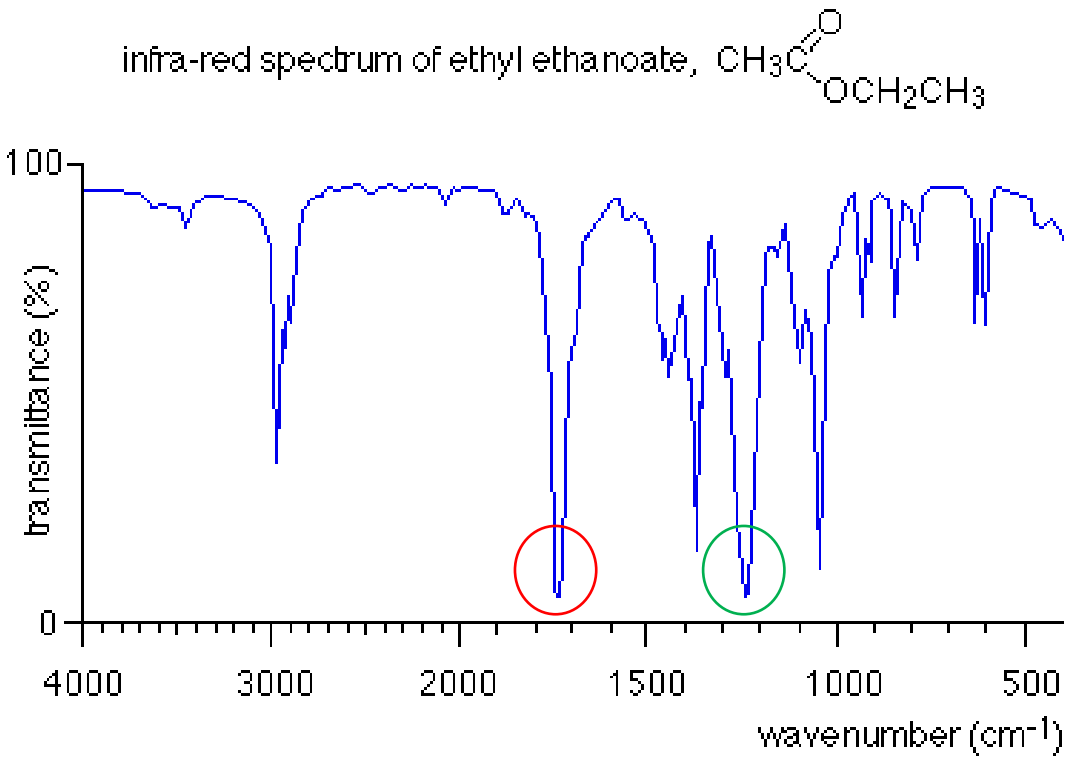
The O-H bond in an alcohol absorbs at a higher wavenumber than it does in an acid - somewhere between 3230 - 3550 $cm^{-1}$.




Notice the absorption due to the C-H bonds just under 3000 $cm^{-1}$, and also the troughs between 1000 and 1100 $cm^{-1}$ - one of which will be due to the C-O bond.

# The infra-red spectrum for an ester

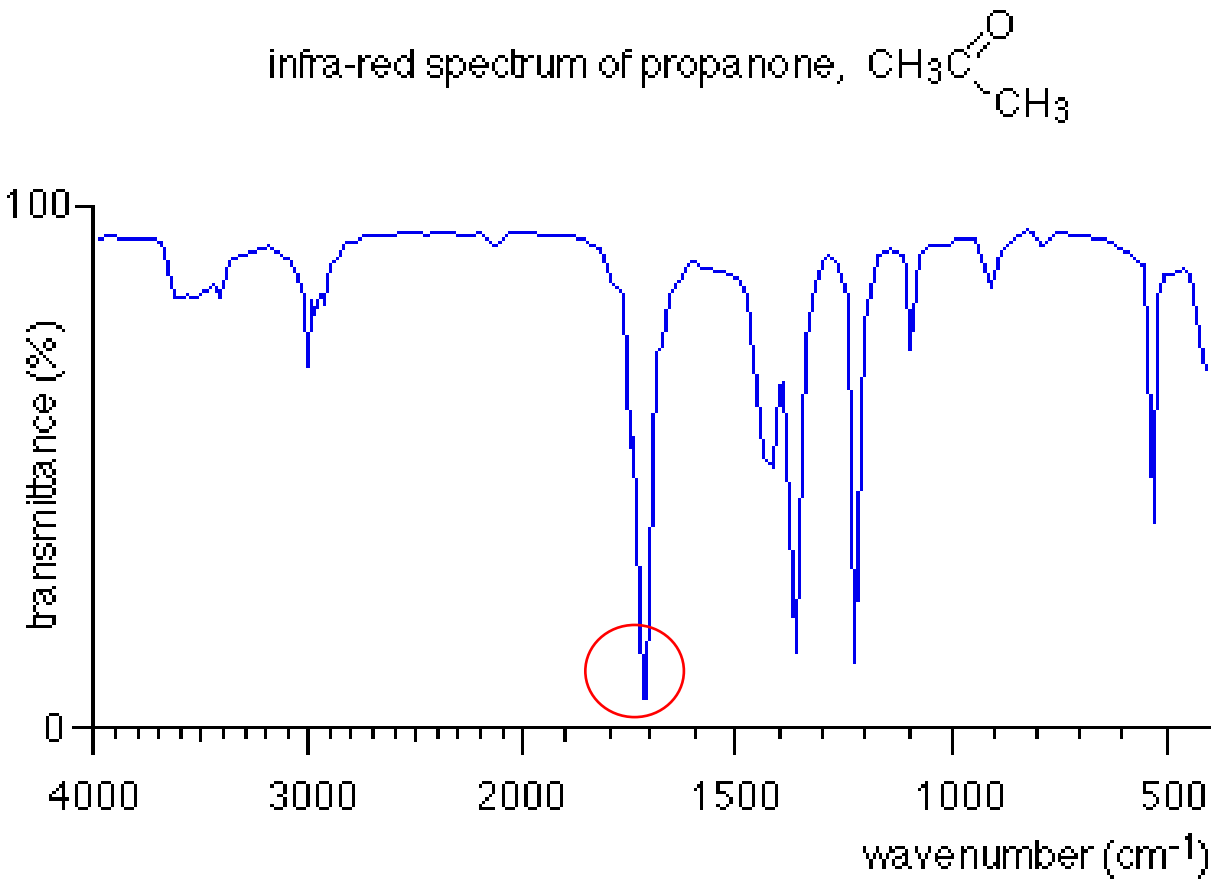
## Ethyl ethanoate




The O-H absorption is missing completely.
The presence of the C=O double bond is seen at about 1740 $cm^{-1}$.
The C-O single bond is the absorption at about 1240 $cm^{-1}$.

# The infra-red spectrum for a ketone

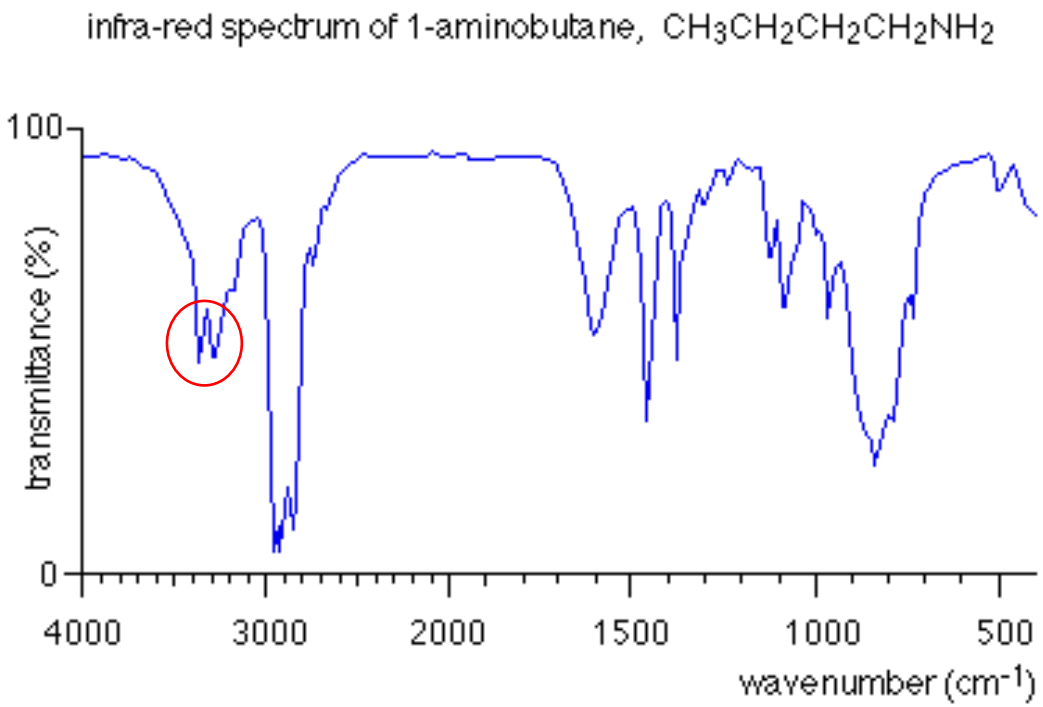
## Propanone




There is no trough due to the O-H bond, and there is a marked absorption at about 1700 $cm^{-1}$ due to the C=O.

# The infra-red spectrum for a primary amine

## 1-aminobutane




Primary amines contain the $-NH_2$ group, and so have N-H bonds.

These absorb somewhere between 3100 and 3500 $cm^{-1}$.

**Double trough (typical of primary amines) can be seen clearly on the spectrum to the left of the C-H absorptions.**